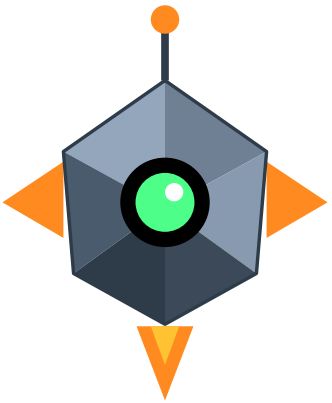
BOOK 3

Two Boxes, One Answer

Boxes that sound right, and boxes that are right





MULTIX // HELLO
I am Multix. I ride along in every chapter. Watch my readout: it shows what is owed, what is paid, and what is only a guess.

CHAPTER 1

The Detective and the Boxes



MULTIX // CIRCLE CHECK
A circle holds a piece that belongs together, and circles can sit inside circles. A circle is *ready* when everything needed for its calculation is known and its operation is allowed by the current rules. To **pop** a ready circle, replace it with its value and leave everything outside it alone. To **redraw**, move circles around or split a number, as long as the value stays the same. The full rules are in the introduction.

STORY

Imagine two plain gray boxes on a table. Each one has a slot on top and a slot on the side. You write a number on a card and drop it in the top, and a moment later a card with a new number slides out of the side.

You cannot open the boxes. You can only feed in numbers and read what comes out. Your job is to work out whether the two boxes are secretly the same.

Here are the rules of the game. Each box follows one fixed rule that never changes. You may drop in any whole number from 0 upward. And you judge a box only by the card that slides out: sounds, smells and how long it takes do not count.

Detectives call this kind of job a *test*. Today you are the detective.

STEP 1 The first test

Drop a 3 into Box A. Out comes a 6. Drop a 3 into Box B. Out comes a 6.

Same answer. Try again.

Put in	Box A gives	Box B gives
1	2	2
2	4	4
5	10	10
7	14	14
12	24	24

Every time, the boxes agree. So are they the same?

Here is the first thing a good detective learns: **agreeing so far is not the same as agreeing forever.** You have only tried five numbers, and there are far more numbers than that.

STEP 2 Looking for the difference

So you keep hunting. You try a big number: 1,000 gives 2,000 in both boxes. You try 0, and both give 0. You try a number you made up that nobody would think of, and the boxes still match.

At some point you start to wonder whether any number can split them.

Here is a secret. Suppose someone tells you the inside of each box.

- Box A takes your number and *multiplies it by 2*.
- Box B takes your number and *adds it to itself*.

Now think about what those two jobs mean. Multiplying a number by 2 means taking two copies of it. Adding a number to itself also means taking two copies of it. They are the same job described in two different ways. Whatever number you put in, there will never be a difference.

So no detective, however clever, can ever tell Box A from Box B using only the slots. The boxes are *different on the inside* but *the same on the outside*. And the outside is all a detective can see.

Trust check. Two different kinds of statements showed up in this chapter.

IN CIRCLES

Think of each box as a circle you cannot open. All you ever see is the number it pops to.

TRACE

```
Box A:  ( 3 × 2 )  →  6
Box B:  ( 3 + 3 )  →  6
```

From outside you see two 6s. Inside, the circles are different. Yet no number can tell the boxes apart, and here is the reason: for *every* n, you can redraw (n + n) as (n × 2) without changing the value, because two copies of n is two copies of n. A redraw that always keeps the value is a reason. A pile of matching pops is only evidence.

STEP 3 The boxes that fooled you

Now meet two new boxes, Box C and Box D. Try them out.

Put in	Box C gives	Box D gives
1	1	1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	8	8
9	9	9

Nine tests in a row, and the boxes matched every time. You are feeling very sure of yourself.

Now try 10.

Box C gives 10. Box D gives 1.

The boxes are different after all!

What was going on inside? Box C simply hands your number back. Box D *adds up the digits* of your number. For a number like 7, there is only one digit, so the sum is 7. But 10 has two digits, 1 and 0, and 1 plus 0 is 1.

Until you tried a two-digit number, there was no way to see the difference. Nine matching tests were not enough.

THE DETECTIVE'S RULE

Write this one on the inside cover of your notebook:

One disagreement proves two boxes are different. When there is no end to the numbers you can drop in, no pile of agreements by itself can prove two boxes are the same.

Why? If a box only accepted the numbers 1 to 10, you could test all ten and be finished. These boxes accept any whole number, so you can never test them all. A single test where the boxes disagree is a fact you can point to. Agreements only tell you that you have not found a difference yet. Maybe there is one hiding at 10, or at 1,000, or at a number you did not think to try.

So when do agreements become a proof? When you can explain *why* the boxes must always agree, the way we did for Box A and Box B. A reason covers every number at once, including all the ones you never tested.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What can the detective see?** (Only the numbers going in and coming out.)
- **What did the detective not see?** (How each box works inside. Box A and Box B work differently, but nobody could tell.)
- **If two boxes always give the same answer, are they the same box?** (They are the same *to anyone using the slots*. They might be built differently on the inside.)

WHAT MAKES THIS CONCLUSION TRUSTWORTHY?

Two different kinds of statements showed up in this chapter. Which is which?

1. "Boxes C and D matched on 1 through 9, so they might be the same."
2. "Boxes A and B must always match, because multiplying by 2 and adding a number to itself are both just two copies."

(Answers: 1 is *evidence*. It is useful, but it can be overturned by one new test. 2 is a *reason*. It covers every number at once.)

PRACTICE

1. Box E adds 1 to your number, then subtracts 1. Box F just hands back your number. Can any input tell them apart? How do you know?

2. Box G multiplies your number by itself. Box H adds your number to itself. Both give 4 when you put in 2. Is that enough to say they are the same? Find an input that splits them.

3. Box I multiplies your number by 5. Box J multiplies your number by 10, then cuts the result in half. Will any number ever tell them apart? Explain your reason in one sentence.

4. **The sneaky box.** Box K adds 1 to your number. Box L also adds 1, except that when you put in exactly 1,000, it gives back 0. You test both boxes on every number from 1 to 999 and they always match. Is it

fair to say they are the same? What would you need to be sure?

5. Box M pops (n × 4). Box N pops ((n + n) + (n + n)). Can any n tell them apart? Use a redraw to explain.

BRIDGE *Bridge to the rest of the series:* The boxes in this chapter are the first example of a question that comes back in every book: what can someone on the outside tell, and what stays hidden? In The Work That Vanishes, 8 times 12 and 10 times 10 minus 2 times 2 are two different expressions that give the same answer, and the cut-and-slide picture is the reason. In A Useful Wrong Answer, a guess and the exact value can look identical until you check at a distance.

PUZZLE PAGE

Spot the Slip

Three notebooks. One is perfect. The other two each hide a single slip. Find it!

TRACE

```
1 ( 37 × 43 )
2 redraw  →  ( ( 40 − 3 ) × ( 40 + 3 ) )
3 redraw  →  ( ( 40 × 40 ) − ( 3 × 3 ) )
4 pop ( 40 × 40 )  →  1,600
5 pop ( 3 × 3 )  →  9
6 pop ( 1,600 − 9 )  →  1,591
```

Notebook A ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 19 × 21 )
2 redraw  →  ( ( 20 − 1 ) × ( 20 + 1 ) )
3 redraw  →  ( ( 20 × 20 ) − ( 1 × ( 1 + 1 ) ) )
4 pop ( 20 × 20 )  →  400
5 pop ( 1 × ( 1 + 1 ) )  →  2
6 pop ( 400 − 2 )  →  398
```

Notebook B ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 64 × 76 )
2 redraw  →  ( ( 70 − 6 ) × ( 70 + 6 ) )
3 redraw  →  ( ( 70 × 70 ) − ( 6 × 6 ) )
4 pop ( 70 × 70 )  →  4,900
5 pop ( 6 × 6 )  →  42
6 pop ( 4,900 − 42 )  →  4,858
```

Notebook C ☐ **perfect** ☐ **slip on line** _____



MULTIX // HINT

A slip can hide in a **pop** (wrong value) or in a **redraw** (value changed). Check each line against the problem at the top.

CHAPTER 2

The Box That Sounds Right

STORY

The boxes from the last chapter had slots and numbers. These ones are different. They are *answer boxes*. You type in a question, and a long, neat, confident explanation slides out.
Today you have two of them, Box P and Box Q, and one question:
What is 48 × 52?

STEP 1 Two answers

Here is what Box P wrote:

> 48 and 52 are both 2 away from 50, so picture a 50 by 50 square with a 2 by 2 corner missing. 50 × 50 = 2,500. The corner is 2 × 2 = 4. So 48 × 52 = 2,500 − 4 = **2,496**.

Here is what Box Q wrote:

> 48 and 52 are both 2 away from 50, so picture a 50 by 50 square with a 2 by 2 corner missing. 50 × 50 = 2,500. The corner is 2 × 2 = 4. So 48 × 52 = 2,500 − 4 = **2,486**.

Read them again. They are almost word for word the same. They are just as tidy, and just as sure of themselves. They differ in a single digit.

Which one is right?

You already know how to find out. Go line by line, the way Kofi's tidy proof worked. Every line is fine until the last one, and there Box Q went wrong: 2,500 minus 4 is 2,496, not 2,486.

Write each answer as a trace, with one line for each pop, and the difference jumps out:

TRACE

```
( ( 50 × 50 ) − ( 2 × 2 ) )
pop ( 50 × 50 )  →  2,500
pop ( 2 × 2 )  →  4
```

```
( 2,500 − 4 )
pop  →  2,496     (Box P)
pop  →  2,486     (Box Q)   this pop is wrong
```

A trace lets you check each pop on its own, instead of trusting the whole answer at once.

Notice something uncomfortable. If you had only seen Box Q's answer, with no Box P to compare it to, nothing about it would have warned you. It was neat. It was confident. It was wrong.

STEP 2 What "slop" means

Trust check. Chapter 1 had two kinds of statements: *evidence* and *reasons*.

People have a word for a certain kind of stuff: **slop**. The word is used for things that *look finished* but show little sign that anyone looked after them properly. It might be an essay, a picture, a comic, a TV episode, a homework page or a set of math answers.

Care can go missing in two places. One is **the check**: nothing shows that anyone made sure it was true. The other is **the design**: nothing shows that anyone took care over what it displays and how loudly. This chapter is about the check. The next chapter is about the design.

Three things are worth knowing about the word.

First, **slop is not about who made it.** A box can make careful, correct work, and a person can make work that nobody checked.

Second, **you only see the page, not the maker's mind.** A wrong answer proves there is an error. It does not prove nobody checked, because a careful maker can still slip. So a good detective says what is missing from the page itself, such as "I can't see a reason for this last line," and leaves the story of how it was made open.

Third, **neatness is cheap.** Anyone, or any box, can make neat paragraphs very fast. Neat paragraphs are not evidence that anyone looked at whether they were true.

That brings us back to the detective's rule from chapter 1: matching on the outside does not prove two things are the same inside. Box P and Box Q look the same from the slot. One of them is right.

STEP 3 Four detective questions

When an answer comes out of a box, a good detective asks four things.

1. **Can I check it a different way?** Try 48 × 52 as 50 × 52

minus 2 × 52. That is 2,600 − 104, which is 2,496. Box P matches. Box Q does not.

2. **Does every line follow from the line before?** Read it slowly. The slip is usually in one line, not spread everywhere.
3. **What would I see if it were wrong?** For a product like this, one easy sign is the last digit: 8 × 2 ends in 6, so the answer must end in 6. 2,486 ends in 6 too, so this test lets Box Q slip past. That is a useful lesson: some checks are weak, and a good detective uses more than one.
4. **Who, or what, checked this before it got to me?** If you cannot tell, treat yourself as the first checker.

None of these questions needs you to be better at math than the box. They only need you to be willing to look.

STEP 4 Being fair to the boxes

It would be a mistake to say "never trust a box." A box that is often right is very useful, the same way a friend who is usually right about directions is useful.

But notice what makes you trust the friend. It is not how confidently they speak. It is that you have checked their directions before, and they were right.

Trust should grow with checks, not with polish.

That is the lesson of Mara's board, too. Her guess about 7/8 inch was useful, but she checked a far-off mark with a long tape before relying on it.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What can you see when an answer box hands you something?** (Only the words. You cannot see whether anything was checked.)
- **What did Box P and Box Q's matching look keep?** (The shape of a good answer.)
- **What did Box Q lose?** (The correct number.)

WHAT MAKES THIS ANSWER TRUSTWORTHY?

Chapter 1 had two kinds of statements: *evidence* and *reasons*. Now there is a third thing that is easy to confuse with them.

1. "Box P and Box Q both sound sure." This is **polish**. It tells you nothing about whether the answer is right.
2. "Box P matches my second method." This is **evidence**. It can still be overturned.
3. "Each line of Box P follows from the one before." This is a **reason**. It covers the whole answer.

(Answers: Polish is not evidence at all. Evidence is useful. A reason is best.)

PRACTICE

For each one, say whether the box is **right** or **wrong**. Then check it a second way.

1. Box says: $28 \times 32 = 30 \times 30 - 2 \times 2 = 900 - 4 = 886$.
2. Box says: $7 \times 13 = 10 \times 10 - 3 \times 3 = 100 - 9 = 91$.
3. Box says: $8 \times 13 = 10 \times 10 - 2 \times 2 = 96$. (Remember the missing corner from The Work That Vanishes.)
4. Find two different ways to check that $15 \times 25 = 375$.
5. A friend copies a neat paragraph about volcanoes into a school report without reading it. A person made this, not a box. Is it still slop? Why or why not?
6. Write the answer to problem 1 as a trace, with one line for each pop. Which pop is wrong?

BRIDGE *Bridge to the rest of the series:* In The Order of Discovery, a tidy proof is valuable because every line can be checked. A tidy answer with no check behind it is where slop hides. In Reversing the Calculation, the calculator kept only the answer and lost how it was made. An answer box does the same, which is why it helps to ask it for the steps and then check them yourself.

PUZZLE PAGE

Color by Answer

Shade every box whose answer is exactly **2,475**. Leave the others blank. What appears?

2468 + 9	2467 + 6	2468 + 4	2487 − 12	2495 − 20	2450 + 28	2 × 1241	2460 + 14
2461 + 10	2451 + 19	2465 + 8	2485 − 10	2478 − 3	95 × 26	2475 + 3	2445 + 20
2477 + 3	5 × 495	2503 − 28	165 × 15	3 × 825	2460 + 15	2508 − 33	2446 + 22
2499 − 24	2438 + 37	2479 + 6	2504 − 29	2441 + 34	2459 + 25	2483 − 8	2495 − 20
2469 + 6	2502 − 27	9 × 275	2491 − 16	2483 − 8	45 × 55	2469 + 6	2449 + 26
2443 + 32	2439 + 36	2470 + 5	2457 + 18	2488 − 13	2469 + 6	2462 + 13	2468 + 7
2468 + 15	2490 − 15	2514 − 39	2454 + 15	2447 + 19	165 × 15	2463 + 12	137 × 18
12 × 206	2495 − 20	2486 − 11	2442 + 27	2466 + 5	90 × 90 − 75 × 75	99 × 25	2464 + 22



MULTIX // SCAN

Some boxes look close and are not. Check the corner on each one.

CHAPTER 3

The Page That Shouts

STORY

Last chapter, Box Q gave a neat, calm, confident answer that was wrong. This chapter turns that around. Here are two pages, and **both of them are right.**
Each page explains why 48 × 52 is 2,496.
Page A shows a big 50 by 50 square. A small 2 by 2 corner is shaded dark. Under the picture are three short lines: *50 × 50 = 2,500. The corner is 2 × 2 = 4. So 48 × 52 = 2,496.*
Page B has a title in six colors: *AMAZING MATH TRICK!!!* Five cartoon mascots are shouting from the edges. Twelve arrows point in different directions. A joke about pizza runs along the bottom. Somewhere in the background is the same square, with the 2 by 2 corner drawn small and in nearly the same color as the paper. The answer, 2,496, is there too, in the same size as everything else.
Every number on both pages is correct. The check from last chapter cannot tell them apart. Yet many readers would find Page B harder to use, and it is a good example of the other kind of slop.

STEP 1 Three kinds of care

A page can be right and still be hard to follow. Making it easy to follow takes care over three things.

Salience means the important thing stands out. When you look at the page, the first thing you notice is the main point.

Simplicity means every part earns its place. Nothing is there just to fill space or to look busy.

Relevance means every part is connected to what the page is about. A joke about pizza has nothing to do with a missing corner.

You can test each one.

- **The squint test (salience).** Squint at the page, or hold it at arm's length. What do you notice first? Is it the main point?
- **The cover test (simplicity).** Cover one part with your hand and ask whether the page is harder to use without

it. If it makes no difference to the job the page has to do, that part may not be earning its place. But ask who the page is for: a reminder, a friendly character or a second explanation can help some readers even when it adds no new fact.

- **The "so what?" test (relevance).** Point at each part and ask: *what does this have to do with the point?* If there is no answer, the part does not belong.

Page A passes all three. The shaded corner is what you see first. Nothing could be covered without losing something. Every line is about the answer.

Page B does badly on all three, for this job. The title is the loudest thing. Most of the decoration could be covered with no loss to a reader who wants the trick. The pizza joke has no connection to the problem.

IN CIRCLES

Imagine each part of a page drawn as a circle. A circle should hold only pieces that belong together, which is relevance. The circle holding the main point should look bigger or bolder than the others, which is salience. A circle that holds nothing the reader needs can be left out, which is simplicity. On Page B, every circle is drawn at the same size and weight, so no circle looks like the main one.

Trust check. Chapter 2 gave three kinds of statements: polish, evidence and reasons.

STEP 2 When everything is loud

Imagine a classroom where every student shouts at the same time. Nobody can hear anyone. Loud only works when most things are quiet.

A page like Page B has the same problem. The colors, mascots and arrows are all at the same volume, so none of them can point at the important thing. The corner, which is the whole point of the trick, gets lost.

There is a word for a page like this: **gaudy**. To the reader, it is too much, too loud, and too much the same.

There is something else to watch for. Some pages work mostly by making you *react*, fast, with big faces, big colors and big claims. Others work by *telling* you something, and they can afford to be quieter, because the something does the work. When a page is all reaction and no point, you can ask what it was grabbing you for.

STEP 3 Why hurry can show

Taking care over salience, simplicity and relevance takes time. Somebody has to decide what the main point is. Somebody

has to cut the extras. Somebody has to ask, for each piece, *does this belong?*

Those are choices, and choices are the first thing to go when something is made in a rush. A comic page finished to meet a deadline, a TV episode made quickly to fill a schedule, a worksheet put together in ten minutes: any of them can skip those choices, and skipping them can leave a page that looks busy instead of clear.

But the reverse does not follow. A busy page does not prove a hurried maker, and a spare page does not prove a careful one. You see the page, not the person. So name what the page does to you, for example "I could not find the main point," and leave the reasons open.

Two things are worth keeping straight.

First, **fast is not the crime.** A quick sketch with one clear point can beat a slow, cluttered painting. Speed is just how care tends to get squeezed out.

Second, **exaggeration is not the crime either.** A map that draws a river wider than it really is, so you can see it, is exaggerating on purpose, and every bit of the exaggeration points at something. A cartoon that stretches the one thing that matters can be careful work. Exaggeration with nothing to point at is just noise.

The honest test is never "who made it?" or "how fast?" It is the three questions: does the main point stand out, does every part earn its place, and does every part belong?

STEP 4 Two questions, four pages

Now we have two questions for any page that comes out of a box, a printer or a person. Is it **right**? Is it **clear for the job it has to do**? Each answer can be yes or no, so there are four kinds of page.

	Clear for this task	Hard to follow
Right	Page A	Page B
Wrong	Box Q's neat answer	A loud page with a wrong answer

You cannot tell which square a page belongs in just by glancing at it. A calm page can be wrong (Box Q). A loud page can be right (Page B). Both questions need a real look.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did Page A and Page B keep in common?** (The correct answer, 2,496.)
- **What did Page B lose?** (Clarity. The reader's attention gets pulled away from the corner.)
- **Could the check from chapter 2 have warned us about Page B?** (No. Page B is right. Its problem is design.)

WHAT MAKES THIS PAGE TRUSTWORTHY?

Chapter 2 gave three kinds of statements: polish, evidence and reasons. This chapter adds a warning. **Being right does not make a page clear, and being clear does not make a page right.** A trustworthy page needs both, and each needs its own look.

PRACTICE

1. **Page C** has the square, the shaded corner and three plain lines of writing, just like Page A. But its last line says $48 \times 52 = 2{,}486$. Is Page C's problem a design problem or a check problem? Which questions would catch it?
2. **Page D** explains the missing corner. It also has a paragraph on the history of squares, a joke about a dog, and a rainbow border. Use the cover test. Which parts would you remove? Is there any part you would keep only if it connected to the point?
3. Think of a comic page, a TV episode or an advert you have seen that felt too busy. Apply the squint test, the cover test and the "so what?" test. What did you find?
4. Fix Page B. Write down what you would delete, what you would make bigger, and what you would make quieter.
5. True or false, and why:
    - (a) "A page made by a person cannot be slop."
    - (b) "A page made quickly is always slop."

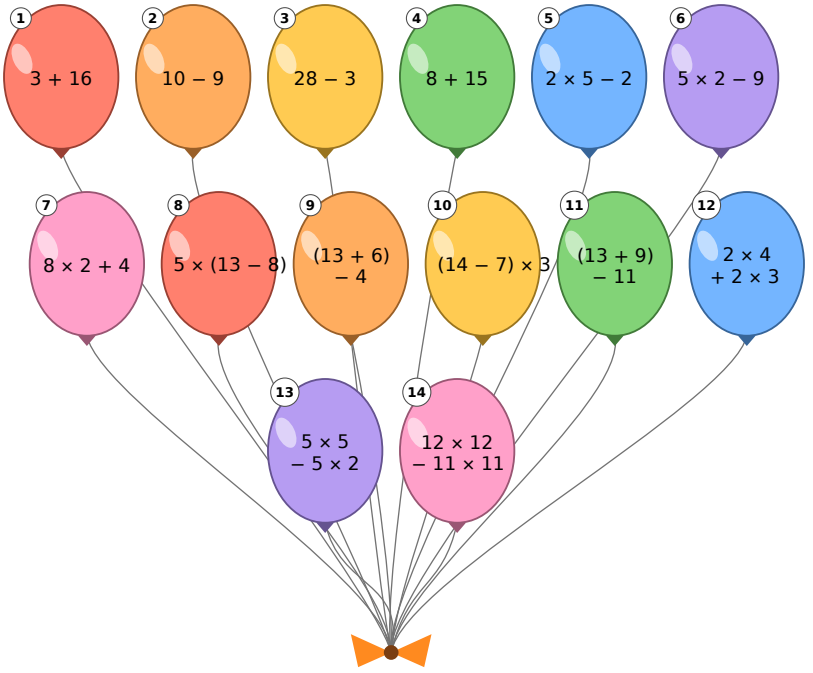
BRIDGE *Bridge to the rest of the series:* In The Work That Vanishes, the whole trick is removing work that was never needed, and a good page does the same. In The Order of Discovery, a tidy proof is tidy because somebody cut out the wrong turns and kept only what belongs. In A Useful Wrong Answer, a good estimate is useful because it names which error matters and which does not, which is relevance again.

PUZZLE PAGE

The Balloon Message

What is Multix's advice? Work out each balloon, use the key to turn the number into a letter, and fill the banner box with the same number.

A = 1	B = 2	C = 3	D = 4	E = 5	F = 6	G = 7	H = 8	I = 9	J = 10	K = 11	L = 12	M = 13
N = 14	O = 15	P = 16	Q = 17	R = 18	S = 19	T = 20	U = 21	V = 22	W = 23	X = 24	Y = 25	Z = 26



What was the answer?

													!
1	2	3	4	5	6	7	8	9	10	11	12	13	14

CHAPTER 4

The Box With a Window

STORY

Here are two new boxes, Box E and Box F. You drop in a card, and a number slides out. Both boxes give **12**, every time.
Box E is built to pop the circle (3 × 4). Box F is built to pop the circle (2 × 6).

STEP 1 Same answer, from outside

From the slot, you only see the 12. So by the detective's rule from chapter 1, you cannot tell Box E from Box F. They agree on everything you can see.

TRACE

```
Box E:  ( 3 × 4 )  →  12
Box F:  ( 2 × 6 )  →  12
```

Now someone cuts a small window in each box. Through the window you can see the notebook, which is the trace of pops. Box E shows a line: *pop (3 × 4) → 12.* Box F shows: *pop (2 × 6) → 12.*

With the window, the boxes are different at a glance. Nothing about their answers changed. What changed is **what you are able to see**.

STEP 2 Which differences can you see?

Here are two more boxes.

Box G always pops the left circle first. **Box H** always pops the right circle first. Both are given the same problem:

TRACE

```
( ( 2 + 3 ) × ( 4 + 1 ) )
```

Here is what each box writes in its window.

TRACE

```
Box G                          Box H
pop ( 2 + 3 )  →  5            pop ( 4 + 1 )  →  5
pop ( 4 + 1 )  →  5            pop ( 2 + 3 )  →  5
pop ( 5 × 5 )  →  25           pop ( 5 × 5 )  →  25
```

Trust check. The reason from chapter 1 still holds, but it now has a limit.

Look through three different windows.

- **The answer-only window.** It shows just 25. The boxes look the same.
- **The pops window.** It shows which circles were popped, as a collection. Both boxes popped the same three circles, so the boxes look the same.
- **The order window.** It shows which circle was popped first. Now Box G and Box H are different.

So whether two boxes are the same depends on which window you have. The same two boxes can be identical through one window and different through another.

STEP 3 Choose the window for the job

Which window is the right one? It depends on who is asking.

A shopper only cares how much she pays. For her, the answer-only window is enough, and two boxes that give the same price are the same.

A cook cares about the recipe. For him, (3 × 4) and (2 × 6) are different, even if both give 12, because they mean different amounts of different things.

A teacher checking homework cares about the order of steps, so the order window matters to her.

There is no single correct window. But there is a correct way to talk: **before you say two boxes are the same, say who is watching and what they can see.**

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What stayed the same when we added a window?** (The answers. Nothing changed inside the boxes.)
- **What changed?** (What the watcher could see, and therefore which boxes counted as the same.)

WHAT MAKES THIS TRUSTWORTHY?

The reason from chapter 1 still holds, but it now has a limit. Redrawing (n + n) as (n × 2) keeps the *value*, so for someone who only sees values, the boxes are always the same. For someone with a pops window, the redraw is a visible change, and the boxes are not the same.
So a statement like “these boxes are the same” is only complete when it names the window. Without that, it is missing a part.

PRACTICE

1. Box J pops (6 + 6). Box K pops (4 + 8). Do they look the same through the answer-only window? Through the pops window?
2. Box L pops ((1 + 2) + (3 + 4)) left circle first. Box M pops it right circle first. Which window tells them apart? What number does each box give?
3. Box N pops (n + 0). Box P pops (n × 1). For every n, do they look the same through the answer-only window? Through the pops window?
4. Name one job where the answer-only window is enough, and one job where you would want to see the pops.
5. (2 × 6) and (12 × 1) both pop to 12. Which window would you need to tell the boxes apart?

BRIDGE *Bridge to the rest of the series:* In Reversing the Calculation, Lena saw only the answer, 12, and could not tell which circle made it. A window on the trace is exactly the clue that would have helped her, and it is the diary from that chapter.

PUZZLE PAGE

Spot the Slip

Three notebooks. One is perfect. The other two each hide a single slip. Find it!

TRACE

```
1 ( 38 × 42 )
2 redraw  →  ( ( 40 − 2 ) × ( 40 + 2 ) )
3 redraw  →  ( ( 40 × 40 ) − ( 2 × 2 ) )
4 pop ( 40 × 40 )  →  1,600
5 pop ( 2 × 2 )  →  6
6 pop ( 1,600 − 6 )  →  1,594
```

Notebook A ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 24 × 26 )
2 redraw  →  ( ( 25 − 1 ) × ( 25 + 1 ) )
3 redraw  →  ( ( 25 × 25 ) − ( 1 × ( 1 + 1 ) ) )
4 pop ( 25 × 25 )  →  625
5 pop ( 1 × ( 1 + 1 ) )  →  2
6 pop ( 625 − 2 )  →  623
```

Notebook B ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 46 × 54 )
2 redraw  →  ( ( 50 − 4 ) × ( 50 + 4 ) )
3 redraw  →  ( ( 50 × 50 ) − ( 4 × 4 ) )
4 pop ( 50 × 50 )  →  2,500
5 pop ( 4 × 4 )  →  16
6 pop ( 2,500 − 16 )  →  2,484
```

Notebook C ☐ **perfect** ☐ **slip on line** _____



MULTIX // HINT

A slip can hide in a **pop** (wrong value) or in a **redraw** (value changed). Check each line against the problem at the top.

CHAPTER 5

Boxes That Agree for a While

STORY

The detective had two new boxes. Box S squares a number: n × n. Box T doubles it: n + n.
She tested them with a card marked 2.

TRACE

```
Box S:  ( 2 × 2 )  →  4
Box T:  ( 2 + 2 )  →  4
```

“Both say 4,” said the detective. “The same box?”
Her helper said, “Try another card.”

STEP 1 One test is not enough

She tried more cards.

Card	Box S (n × n)	Box T (n + n)
1	1	2
2	4	4
3	9	6
4	16	8

The boxes agree at 2 only (and at 0, if you try that). At any other card, they differ. The test with card 2 was a coincidence.

This is the detective’s rule from chapter 1 in action: two boxes agreeing on the cards you tried does not mean they agree on every card.

STEP 2 A pair that really is the same

Now two other boxes. Box U squares a number: n × n. Box V does something odd: it takes one less than n, multiplies by one more than n, then adds 1.

TRACE

```
Box V:  ( ( ( n − 1 ) × ( n + 1 ) ) + 1 )
```

Test them with cards 1 to 5. (The cards start at 1, so that n − 1 is never below 0.)

Trust check. A test can show two boxes are different, with certainty: one card where they disagree is enough.

Card	Box U	Box V
1	1	1
2	4	4
3	9	9
4	16	16
5	25	25

Five cards, five agreements. Should the detective trust it now?

Not yet. But now she can look for a **reason**, and there is a good one from The Work That Vanishes. The product $(n - 1) \times (n + 1)$ is a square with its corner missing:

TRACE

$$(n - 1) \times (n + 1) = (n \times n) - (1 \times 1)$$

So Box V is $(n \times n)$ minus 1, plus 1, which is just $n \times n$ again. The boxes are the same for **every** card, because the missing corner and the +1 cancel exactly.

STEP 3 Two kinds of agreement

Compare the two pairs.

- Boxes S and T agreed on one card and had **no reason** to agree anywhere else.
- Boxes U and V agreed on five cards and have a **reason** that holds for all of them.

The number of agreeing cards was not the difference. The reason was.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the agreeing cards keep?** (A list of cases that match.)
- **What did the reason keep?** (A promise about every case, including those not tried.)

WHAT MAKES THIS TRUSTWORTHY?

A test can show two boxes are different, with certainty: one card where they disagree is enough. A test can never show two boxes are the same for all cards. Only a reason can, such as the redraw above, which keeps the value for every n.

PRACTICE

1. Box W triples a number: (n + n) + n. Box X multiplies by 3: (3 × n). Do they agree for every card? Give a reason.
2. Box S squares a number and Box Y multiplies it by 4: (4 × n). Find a card where they agree besides 0.
3. Box S and Box Y disagree at card 5. What do they give?
4. A detective tests two boxes on cards 1, 2, 3, 4, 5, 6, 7, 8, 9 and 10 and they always agree. Can she say they are the same? What would she need?
5. Use the missing-corner trick to explain why (9 × 11) + 1 equals (10 × 10).

BRIDGE *Bridge to the rest of the series:* In The Order of Discovery, a pattern can hold for forty examples and then fail. Here two boxes can agree on one card and then differ, and a reason is what separates a coincidence from a rule.

PUZZLE PAGE

Color by Answer

Shade every box whose answer is exactly **2,496**. Leave the others blank. What appears?

2471 + 20	2491 + 11	2490 + 2	2501 − 5	2531 − 35	2498 + 6	2477 + 10	2487 + 5
2470 + 28	35 × 71	96 × 96 − 82 × 82	61 × 61 − 35 × 35	2529 − 33	2470 + 22	2471 + 23	2485 + 16
2490 + 2	2486 + 10	39 × 64	2472 + 24	4 × 624	2471 + 25	2 × 1248	2502 + 3
32 × 78	2509 − 13	2493 + 5	2514 − 18	2476 + 20	2475 + 28	55 × 55 − 23 × 23	2478 + 18
2482 + 14	110 × 110 − 98 × 98	2466 + 30	2464 + 32	2523 − 27	2519 − 23	156 × 16	2493 + 3
2513 − 17	2483 + 13	2479 + 17	2509 − 13	2514 − 18	2467 + 29	2526 − 30	2533 − 37
2497 + 5	2522 − 26	32 × 78	2460 + 29	2494 + 9	4 × 624	2474 + 22	2470 + 18
2478 + 8	2514 − 18	52 × 48	2488 + 16	2477 + 16	2461 + 35	2480 + 16	2496 + 12



MULTIX // SCAN

Some boxes look close and are not. Check the corner on each one.

CHAPTER 6

The Box With a Secret Exception

STORY

The detective was given two new boxes. Box A doubles a card: (n × 2). Box B looks like it does the same. She tested six cards.

Card	Box A	Box B
1	2	2
2	4	4
3	6	6
4	8	8
5	10	10
6	12	12

Six agreements. The detective was about to write “same box”. Then her helper opened a small window in Box B.

STEP 1 The rule inside

Through the window, she could read Box B’s rule:

TRACE

```
if the card is 7:   give 0
otherwise:          ( n × 2 )
```

Box B has a **secret exception** at card 7. At every other card it matches Box A, but not there:

TRACE

```
Box A at 7:  ( 7 × 2 )  →  14
Box B at 7:  0
```

The detective tested cards 1 to 6 and never touched card 7. No amount of testing the same kind of cards would have found it, unless she happened to try 7.

STEP 2 What testing can and cannot do

- If the detective tries a card where the boxes differ, she **knows** they are different.

Trust check. Looking inside is trustworthy when the window shows the real rule.

- If she never tries that card, she learns nothing about it.

The exception was a single card out of endlessly many. To find it by testing alone, she would have to be lucky, or try every card.

If the cards were only the numbers 1 to 10, she could try all ten. That is a complete test, and then agreement would really mean “same box”. With endlessly many cards, no list of tests can ever be complete.

STEP 3 The window is the answer

The window did what testing could not. It showed **how the box is built**, so the detective could read the exception directly.

This is the idea from chapter 4: what you can see decides which boxes look the same. With a window on the rule, Box B and Box A look different at once. With only the slot, they look the same on six cards.

A reason about how the boxes are built covers every card at once, including the ones nobody tries. That is why chapter 5 looked for a reason, not more tests.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the six tests keep?** (Six matching cards.)
- **What did they miss?** (The exception at card 7.)

WHAT MAKES THIS TRUSTWORTHY?

Looking inside is trustworthy when the window shows the real rule. A claim like “the boxes are the same” can rest on a reason about their rules, or on a complete test of every card. It cannot rest on a partial test.

PRACTICE

1. Box C doubles every card, except it gives 100 for card 3. Box D doubles every card. At which card do they differ? What does each give?
2. The detective tests Box C and Box D on cards 1, 2, 4 and 5. What does she see? Is she right to say they are the same?
3. The cards are only 1, 2, 3 and 4. How many tests

settle whether Box C and Box D agree on every card?

4. Why does a window help more than more tests?

5. Box E gives (n × 2) for every card except 1,000,000, where it gives 0. About how many cards would you test before you found the exception, if you tried 1, 2, 3 and so on?

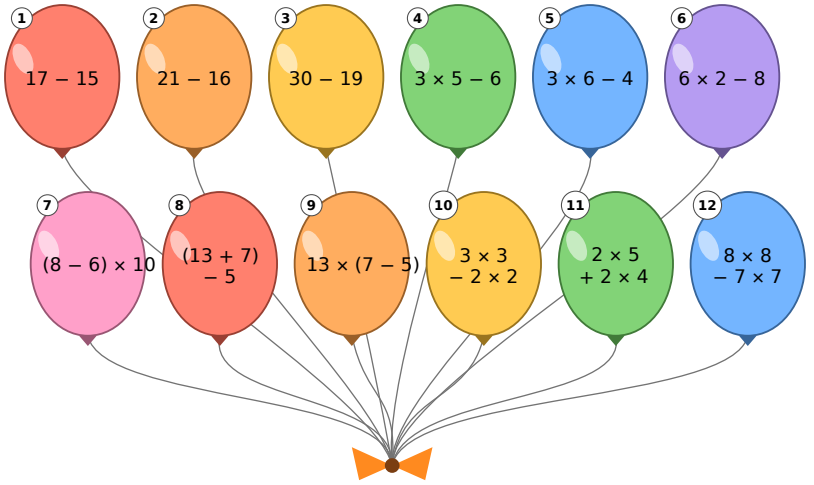
BRIDGE *Bridge to the rest of the series:* In The Order of Discovery, a pattern held for forty examples and then failed. Here the failure is hidden at one card. In Reversing the Calculation, a trace is a window onto the steps, and it shows what a bare answer cannot.

PUZZLE PAGE

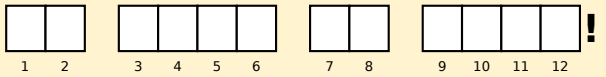
The Balloon Message

What is Multix's advice? Work out each balloon, use the key to turn the number into a letter, and fill the banner box with the same number.

A = 1	B = 2	C = 3	D = 4	E = 5	F = 6	G = 7	H = 8	I = 9	J = 10	K = 11	L = 12	M = 13
N = 14	O = 15	P = 16	Q = 17	R = 18	S = 19	T = 20	U = 21	V = 22	W = 23	X = 24	Y = 25	Z = 26



What was the answer?



CHAPTER 7

Same Answer, Different Cost

STORY

The detective had two boxes that both multiply 7 by a card. She dropped in the card 10.

TRACE

```
Box M:  7 + 7 + 7 + 7 + 7 + 7 + 7 + 7 + 7 + 7  →  70
Box N:  ( 7 × 10 )  →  70
```

Both give 70. By the answer-only window, they are the same box.
But she opened the cost window, which counts the pops.

STEP 1 Count the pops

Box M adds one seven at a time. To add ten sevens, it does 9 additions.

TRACE

```
( 7 + 7 )  →  14
( 14 + 7 )  →  21
( 21 + 7 )  →  28
...
( 63 + 7 )  →  70
```

Box N does one multiplication. That is **1 pop**.

Nine pops against one pop. Same answer, very different cost.

STEP 2 Bigger cards, bigger gap

Now drop in the card 1,000.

- Box M adds 7 a thousand times: **999 additions**.
- Box N does (7 × 1,000) → 7,000: **1 pop**.

Box M's cost grows with the card. Box N's cost stays the same. If a box has to answer a million cards, that difference matters a great deal.

Trust check. A claim about boxes is trustworthy when it names the window.

STEP 3 Windows again

This is the same idea as chapter 4: whether two boxes are the same depends on the window.

- **Answer-only window:** M and N are the same.
- **Pops window:** M and N are different. They popped different circles.
- **Cost window:** M and N are very different. One is cheap and the other grows.

A shopper at the till only needs the answer, but a shop that serves thousands of customers cares about the cost. There is no single right window. The right one depends on what you need.

STEP 4 Same cost does not mean same box

The reverse also holds. Two boxes can cost the same and still be different boxes.

TRACE

```
Box P:  ( n + n )
Box Q:  ( n × n )
```

Each takes one pop. But at card 5, Box P gives 10 and Box Q gives 25. Equal cost, different answers.

So “same answer” and “same cost” are two different questions. A pair of boxes can match on one, on both, or on neither.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did M and N keep in common?** (The answer, for every card.)
- **What did the cost window show?** (A difference the answer-only window could not.)

WHAT MAKES THIS TRUSTWORTHY?

A claim about boxes is trustworthy when it names the window. “M and N are the same” is true in the answer window. “M and N are equally good” is false in the cost window. If someone says two things are the same, ask: the same in what way?

PRACTICE

1. Box M adds 7 twelve times. How many additions is that? What does Box N do instead?
2. Box R adds 1 to a card, ten times in a row. Box S adds

10 once. Do they give the same answer? Count the pops.

3. Box P gives (n + n) and Box Q gives (n × 2). Do they agree on every card? What is their cost?

4. Which window would a shopper at a till use? Which would a shop owner use?

5. Box T takes 1 pop and Box U takes 1 pop but they disagree at card 5. What does that show about cost and answers?

BRIDGE *Bridge to the rest of the series:* In The Boundary of a Step, the number of steps depends on the menu and on the idea behind it. Here the cost window is a new way of looking at a box. In The Order of Discovery, the pairing idea made a hundred additions into two pops.

PUZZLE PAGE

Spot the Slip

Three notebooks. One is perfect. The other two each hide a single slip. Find it!

TRACE

```
1 ( 37 × 43 )
2 redraw  →  ( ( 40 − 3 ) × ( 40 + 3 ) )
3 redraw  →  ( ( 40 × 40 ) − ( 3 × 3 ) )
4 pop ( 40 × 40 )  →  1,600
5 pop ( 3 × 3 )  →  9
6 pop ( 1,600 − 9 )  →  1,591
```

Notebook A ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 19 × 21 )
2 redraw  →  ( ( 20 − 1 ) × ( 20 + 1 ) )
3 redraw  →  ( ( 20 × 20 ) − ( 1 × 1 ) )
4 pop ( 20 × 20 )  →  400
5 pop ( 1 × 1 )  →  2
6 pop ( 400 − 2 )  →  398
```

Notebook B ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 64 × 76 )
2 redraw  →  ( ( 70 − 6 ) × ( 70 + 6 ) )
3 redraw  →  ( ( 70 × 70 ) − ( 6 × ( 6 + 1 ) ) )
4 pop ( 70 × 70 )  →  4,900
5 pop ( 6 × ( 6 + 1 ) )  →  42
6 pop ( 4,900 − 42 )  →  4,858
```

Notebook C ☐ **perfect** ☐ **slip on line** _____



MULTIX // HINT

A slip can hide in a **pop** (wrong value) or in a **redraw** (value changed). Check each line against the problem at the top.

TWO BOXES, ONE ANSWER

CHAPTER 8

Boxes in a Row

STORY

The detective put two boxes in a row so that the output of the first went straight into the second.

- Box A adds 3 to a card.
- Box B doubles a card.

She tried two orders.

STEP 1 A then B, and B then A

A first, then B. Add 3, then double.

TRACE

```
card 5:  ( 5 + 3 )  →  8,   ( 8 × 2 )  →  16
```

B first, then A. Double, then add 3.

TRACE

```
card 5:  ( 5 × 2 )  →  10,   ( 10 + 3 )  →  13
```

The same boxes, in a different order, give different answers: 16 against 13. The order of boxes matters.

STEP 2 Why they differ

With a number n on the card, the circles show it.

TRACE

```
A then B:   ( ( n + 3 ) × 2 )   →   ( ( n × 2 ) + 6 )
B then A:   ( ( n × 2 ) + 3 )
```

The first adds 6 and the second adds 3. They differ by 3 for every card. So these two rows never agree, however many cards you try.

Trust check. The claim “these rows are the same” rests on the redraws, which hold for every card.

STEP 3 Rows that do not care about order

Now Box C, which adds 4. Try A then C and C then A.

TRACE

```
A then C:   ( ( n + 3 ) + 4 )   →   ( n + 7 )
C then A:   ( ( n + 4 ) + 3 )   →   ( n + 7 )
```

Same answer, either way. Adding 3 and adding 4 do not interfere with each other.

Another pair: Box B doubles, Box D triples.

TRACE

```
B then D:   ( ( n × 2 ) × 3 )   →   ( n × 6 )
D then B:   ( ( n × 3 ) × 2 )   →   ( n × 6 )
```

Same again. Doubling and tripling do not interfere.

So some boxes can swap places and some cannot. Adding and doubling cannot swap. Adding and adding can. Doubling and tripling can.

STEP 4 Two different rows that agree

Here is an unexpected pair. Compare:

- **Row 1:** add 3, then double.
- **Row 2:** double, then add 6.

TRACE

```
Row 1:   ( ( n + 3 ) × 2 )   →   ( ( n × 2 ) + 6 )
Row 2:   ( ( n × 2 ) + 6 )
```

They give the same answer for every card, though they are built differently. Card 5: Row 1 gives 16 and Row 2 gives (5 × 2) + 6 = 16. The reason is a redraw: sharing the 2 with both parts of (n + 3).

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did swapping the boxes change?** (The answer, when the boxes interfere with each other.)
- **What did the redraw show?** (Why two different rows can still be the same.)

WHAT MAKES THIS TRUSTWORTHY?

The claim “these rows are the same” rests on the redraws, which hold for every card. The claim “these rows are different” rests on a single card where the answers disagree. As in earlier chapters, one case can break a claim, but only a reason can prove it.

PRACTICE

1. Box E adds 5 and Box F adds 8. Does the order matter? Show with circles.
2. Box G doubles and Box H subtracts 2. Try both orders on the card 7.
3. Is “subtract 2, then double” the same as “double, then subtract 4”? Give a reason.
4. Box J halves a card and Box K adds 1. Try both orders on the card 10.
5. Name two boxes whose order never matters.

BRIDGE *Bridge to the rest of the series:* In chapter 4, we saw that an order window can tell boxes apart. Here the order of the boxes themselves is what changes the answer. In Reversing the Calculation, undoing a chain of buttons in the opposite order is why socks and shoes come off in reverse.
Next in this book: Chapter 9: The Answer on the Fridge.

PUZZLE PAGE

Color by Answer

Shade every box whose answer is exactly **3,599**. Leave the others blank. What appears?

3573 + 23	3570 + 25	3584 + 18	3608 − 9	3610 − 11	3588 + 8	3578 + 17	3572 + 28
3585 + 7	3580 + 11	5 × 722	3617 − 18	3635 − 36	3585 + 26	3581 + 8	3572 + 25
3572 + 22	3634 − 35	3571 + 28	3571 + 28	3575 + 24	3567 + 32	3621 − 22	3577 + 17
3623 − 24	3561 + 38	3595 + 16	3613 − 14	3616 − 17	3590 + 6	3593 + 6	3622 − 23
3634 − 35	3564 + 35	3592 + 7	3614 − 15	3604 − 5	3613 − 14	3628 − 29	3630 − 31
3595 + 4	3584 + 15	3587 + 12	3619 − 20	3589 + 10	3605 − 6	3627 − 28	3627 − 28
3571 + 18	3566 + 33	3624 − 25	3591 + 18	3577 + 29	3607 − 8	3570 + 29	3592 + 9
2 × 1797	3634 − 35	3617 − 18	3593 + 7	3577 + 26	3606 − 7	3580 + 19	3592 + 17



MULTIX // SCAN

Some boxes look close and are not. Check the corner on each one.

CHAPTER 9

The Answer on the Fridge

STORY

Lena's class was making three identical quilts. Each quilt had 48 rows with 52 squares in each row. The question on the card was: **how many squares in all three quilts?** Two boxes were given the same job.

STEP 1 The first box does everything again

The first box wrote the sum, and then worked out each quilt from the beginning.

TRACE

```
( ( 48 × 52 ) + ( 48 × 52 ) + ( 48 × 52 ) )
```

For each quilt, it used the missing-corner trace from the case file. Each one took five lines: two redraws and three pops.

TRACE

```
redraw  →  ( ( 50 − 2 ) × ( 50 + 2 ) )
redraw  →  ( ( 50 × 50 ) − ( 2 × 2 ) )
pop ( 50 × 50 )  →  2,500
pop ( 2 × 2 )  →  4
pop ( 2,500 − 4 )  →  2,496
```

Three quilts is 15 lines. Then it needed two more pops to add the three answers: (2,496 + 2,496) and then (4,992 + 2,496). The total is 17 lines and the answer is **7,488**.

STEP 2 The second box writes a note

The second box noticed that the three circles were exactly the same circle. They would all give the same value. So it did the work **once**, and wrote the answer on a sticky note on the fridge.

Trust check. The second box is trustworthy because its note is checked, not assumed.

TRACE

```
pop ( 48 × 52 )  →  2,496      (five lines of work, then a note)
reuse note  →  2,496
reuse note  →  2,496
pop ( 2,496 + 2,496 )  →  4,992
pop ( 4,992 + 2,496 )  →  7,488
```

The first circle took the usual five lines. A **reuse** is a pop whose answer comes from the notebook instead of from fresh work. The trace still gets a line for it, so a reader can see where the value came from. That is 5 + 2 + 2 = 9 lines.

Both boxes give **7,488**. The second box took 9 lines, and the first took 17.

STEP 3 When the note is safe to use

A sticky note only helps if the circles really are the same. Lena checked a different job: one quilt of 48 × 52 and one of 48 × 53.

TRACE

```
( 48 × 52 )  and  ( 48 × 53 )
```

These are different circles. The note that says 2,496 belongs to 48 × 52 only. Using it for 48 × 53 would be a wrong reuse. The second one is 2,544, as you saw in The Work That Vanishes.

What about **52 × 48**? It is a different way of writing the same product, so the note applies. Lena knows this because the order of the two numbers does not change a product. She writes why on the note.

STEP 4 What the note costs

Reuse is not free. Someone has to write the note, keep it, and check that the circle matches. For three quilts, the saving is big. For a circle you only need once, a note costs more than it saves.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the second box keep?** (The same answer, 7,488.)
- **What did it change?** (How much work it did: 9 lines instead of 17.)

WHAT MAKES THIS TRUSTWORTHY?

The second box is trustworthy because its note is checked, not assumed. The reuse is allowed only when the two circles are exactly the same, and the reason is written down. Both boxes get the same answer, so a reader can compare their traces and see where the saving came from.

PRACTICE

1. How many lines would the first box take for **four** quilts of 48 × 52, if each quilt takes 5 lines and each addition is one pop? How many for the second box?

2. For each pair, say whether one note can serve both: (a) 48 × 52 and 52 × 48, (b) 48 × 52 and 49 × 51, (c) 50 × 50 and 50 × 50.

3. Lena needs 49 × 51 twice. Use the corner trick once, write the note, and find the total.

4. A note says 2,496. Someone wants to use it for 48 × 52 + 1. Is that a reuse of the note? What do they still have to do?

5. A circle will be used only one time. Should the second box bother with a note? Why?

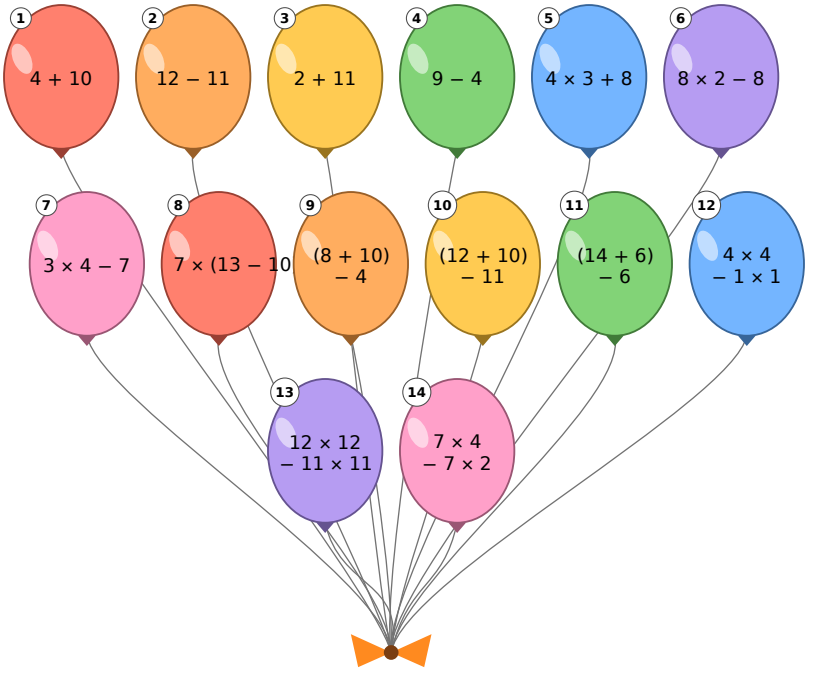
BRIDGE *Bridge to the rest of the series:* In The Boundary of a Step, the cost of a check and the cost of an idea were counted. A note on the fridge is another cost to count. In The Order of Discovery, three different routes to 48 × 52 all reach the same note.

PUZZLE PAGE

The Balloon Message

What is Multix's advice? Work out each balloon, use the key to turn the number into a letter, and fill the banner box with the same number.

A = 1	B = 2	C = 3	D = 4	E = 5	F = 6	G = 7	H = 8	I = 9	J = 10	K = 11	L = 12	M = 13
N = 14	O = 15	P = 16	Q = 17	R = 18	S = 19	T = 20	U = 21	V = 22	W = 23	X = 24	Y = 25	Z = 26



What was the answer?

													!
1	2	3	4	5	6	7	8	9	10	11	12	13	14

Answers (turn the book around)

Chapter 1

1. No input can tell them apart. Adding 1 and then subtracting 1 puts you right back where you started, whatever number you used. That is a reason, not just a pile of tests.
2. Matching on 2 is not enough. Put in 3: Box G gives 9 and Box H gives 6, so they are different. (The only numbers where they agree are 0 and 2, which is why 2 was a trap.)
3. No. Ten copies of a number cut in half is five copies of it. That works for every number.
4. No, it is not fair. Nine hundred and ninety-nine matching tests still leave 1,000 untested, and 1,000 is where they split. To be sure, you would need either to test 1,000 or to see inside the boxes. Because these boxes accept any whole number, testing alone can never cover them all, so a reason about how the boxes are built is the way to be completely sure.
5. No n can tell them apart. ((n + n) + (n + n)) is four copies of n, so it redraws as (n × 4) for every n, and the value never changes.

Spot the Slip.
Notebook A: **perfect**. 37 × 43 = 1,591.
Notebook B: slip on line **3**. The corner is 1 × 1, not 1 × (1 + 1). The answer should be 399.
Notebook C: slip on line **5**. 6 × 6 is 36. The answer should be 4,864.

Chapter 2

1. **Wrong.** 900 − 4 is 896, not 886. Second check: 28 × 32 is 28 × 30 + 28 × 2 = 840 + 56 = 896.
2. **Right.** 7 and 13 are both 3 away from 10, and 100 − 9 is 91. Second check: 7 × 13 is 7 × 10 + 7 × 3 = 70 + 21 = 91.
3. **Wrong.** 8 and 13 are not the same distance from a whole-number middle, so the 2 by 2 corner does not apply. The real answer is 8 × 12 + 8 = 104.
4. One way is 20 × 20 − 5 × 5 = 400 − 25 = 375. Another is 15 × 25 = 15 × 100 ÷ 4 = 1,500 ÷ 4 = 375. A third is 15 × 20 + 15 × 5 = 300 + 75 = 375.
5. **It can be.** The word is not about who made it. If the friend really copied the paragraph without reading it, nothing in the report shows that anyone looked at whether it was true or whether it answered the question. A detective would ask the friend rather than assume.
6. ((30 × 30) − (2 × 2)): pop to 900, pop to 4, and then (900 − 4) should pop to 896. The last pop, which gave 886, is the wrong one.

Color by Answer (2,475). The shaded boxes make a robot.



Answers (turn the book around)

Chapter 3

1. It is a **check problem**. The design is clean, but the last line is wrong. The second-method check ($50 \times 52 - 2 \times 52 = 2{,}496$) would catch it.
2. The dog joke and the rainbow border fail the cover test, since the page is no worse without them. The history paragraph stays only if it helps explain the trick. If it wanders, it goes too.
3. Open answer. A good answer names one specific part that failed a test, for example "the background characters could be covered with no loss."
4. Delete the mascots, the arrows and the pizza joke. Make the 2 by 2 corner large and give it a color that stands out from the paper. Shrink the title and make it one color. Keep the three-line explanation.
5. (a) **False.** A person can make a page that is unchecked or hard to follow, just as a box can. (b) **False.** A quick sketch with a clear point is not slop. The word is about how a page serves its reader, not how fast it was made.

Balloon Message.

1.	$3 + 16$	$= 19$	S
2.	$10 - 9$	$= 1$	A
3.	$28 - 3$	$= 25$	Y
4.	$8 + 15$	$= 23$	W
5.	$2 \times 5 - 2$	$= 8$	H
6.	$5 \times 2 - 9$	$= 1$	A
7.	$8 \times 2 + 4$	$= 20$	T
8.	$5 \times (13 - 8)$	$= 25$	Y
9.	$(13 + 6) - 4$	$= 15$	O
10.	$(14 - 7) \times 3$	$= 21$	U
11.	$(13 + 9) - 11$	$= 11$	K
12.	$2 \times 4 + 2 \times 3$	$= 14$	N
13.	$5 \times 5 - 5 \times 2$	$= 15$	O
14.	$12 \times 12 - 11 \times 11$	$= 23$	W

Message: **SAY WHAT YOU KNOW!**

Chapter 4

1. Through the answer-only window, both show 12, so they look the same. Through the pops window they differ: one popped (6 + 6) and the other popped (4 + 8).
2. The **order window** tells them apart. Both boxes give **10**, since (1 + 2) is 3, (3 + 4) is 7, and 3 + 7 is 10.
3. Through the answer-only window they look the same for every n, because adding 0 and multiplying by 1 both give back n. Through the pops window they are different, because one box adds and the other multiplies.
4. Many answers work. For example: checking the price on a bill needs only the answer. Checking someone's homework wants the pops.
5. The pops window. Box (2×6) pops that circle and box (12×1) pops a different one.

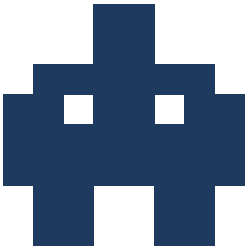
Spot the Slip.
Notebook A: slip on line **5**. 2×2 is 4. The answer should be 1,596.
Notebook B: slip on line **3**. The corner is 1×1, not $1 \times (1 + 1)$. The answer should be 624.
Notebook C: **perfect**. $46 \times 54 = 2{,}484$.

Answers (turn the book around)

Chapter 5

1. Yes. Three lots of n, whether written (n + n) + n or (3 × n), are the same. The reason is that each is n taken three times.
2. Card **4** works: 4 × 4 = 16 and 4 × 4 = 16.
3. Box S gives 25, Box Y gives 20.
4. Not yet. Ten agreements are evidence, not a promise about every card. She needs a reason, such as a redraw that turns one box into the other.
5. (9 × 11) is a 10 by 10 square with a 1 by 1 corner missing, so it is 100 − 1 = 99. Adding the 1 back gives 100.

Color by Answer (2,496). The shaded boxes make a robot.



Chapter 6

1. They differ at card **3**. Box D gives 6 and Box C gives 100.
2. She sees 2, 4, 8 and 10 from both boxes. She is not right to say they are the same, because she skipped card 3.
3. Four tests, one for each card. Then agreement would mean the boxes are really the same on these cards.
4. A window shows the rule for every card at once. Each test only shows one card.
5. About **a million**. You would have to reach card 1,000,000 before you saw the difference.

Balloon Message.

1.	17 − 15	= 2	B
2.	21 − 16	= 5	E
3.	30 − 19	= 11	K
4.	3 × 5 − 6	= 9	I
5.	3 × 6 − 4	= 14	N
6.	6 × 2 − 8	= 4	D
7.	(8 − 6) × 10	= 20	T
8.	(13 + 7) − 5	= 15	O
9.	13 × (7 − 5)	= 26	Z
10.	3 × 3 − 2 × 2	= 5	E
11.	2 × 5 + 2 × 4	= 18	R
12.	8 × 8 − 7 × 7	= 15	O

Message: **BE KIND TO ZERO!**

Answers (turn the book around)

Chapter 7

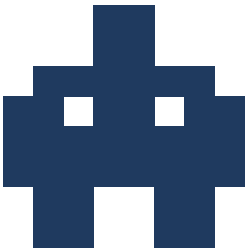
1. Twelve sevens take **11 additions**. Box N does (7 × 12) → 84, which is **1 pop**.
2. Yes, both add 10 to the card. Box R takes **10 pops** and Box S takes **1 pop**.
3. Yes, they agree on every card, because n + n is two lots of n. Each takes 1 pop.
4. A shopper uses the answer-only window. A shop owner also wants the cost window.
5. Equal cost does not mean equal answers. The boxes are different even though they cost the same.

Spot the Slip.
Notebook A: **perfect**. 37 × 43 = 1,591.
Notebook B: slip on line **5**. 1 × 1 is 1. The answer should be 399.
Notebook C: slip on line **3**. The corner is 6 × 6, not 6 × (6 + 1). The answer should be 4,864.

Chapter 8

1. ((n + 5) + 8) and ((n + 8) + 5) both come to (n + 13), so the order does not matter.
2. Double then subtract 2: 14 − 2 = **12**. Subtract 2 then double: 5 × 2 = **10**. The order matters.
3. Yes. ((n − 2) × 2) redraws to ((n × 2) − 4). The 2 is shared with both parts.
4. Halve then add 1: 5 + 1 = **6**. Add 1 then halve: 11 ÷ 2 = **5.5**. The order matters.
5. For example “add 3” and “add 4”, or “double” and “triple”.

Color by Answer (3,599). The shaded boxes make a robot.



Answers (turn the book around)

Chapter 9

1. First box: $4 \times 5 = 20$ lines for the quilts, plus 3 pops to add four answers: **23** lines. Second box: 5 lines for the first quilt, 3 reuses, and 3 additions: **11** lines.
2. (a) Yes, since the order of a product does not matter. (b) No, they are different circles: 2,496 and 2,499. (c) Yes, they are the same circle.
3. $49 \times 51 = 2{,}500 - 1 = 2{,}499$. Using it twice, $2{,}499 + 2{,}499 =$ **4,998**.
4. The note covers only the 48×52 part. They still have to add the 1: $2{,}496 + 1 =$ **2,497**.
5. No. A note costs effort to write and check, and a circle used once gets nothing back.

Balloon Message.

1.	$4 + 10$	$= 14$	N
2.	$12 - 11$	$= 1$	A
3.	$2 + 11$	$= 13$	M
4.	$9 - 4$	$= 5$	E
5.	$4 \times 3 + 8$	$= 20$	T
6.	$8 \times 2 - 8$	$= 8$	H
7.	$3 \times 4 - 7$	$= 5$	E
8.	$7 \times (13 - 10)$	$= 21$	U
9.	$(8 + 10) - 4$	$= 14$	N
10.	$(12 + 10) - 11$	$= 11$	K
11.	$(14 + 6) - 6$	$= 14$	N
12.	$4 \times 4 - 1 \times 1$	$= 15$	O
13.	$12 \times 12 - 11 \times 11$	$= 23$	W
14.	$7 \times 4 - 7 \times 2$	$= 14$	N

Message: **NAME THE UNKNOWN!**